RxRosetta

A post discharge medication companion that turns confusing hospital paperwork into a safe, verified, multilingual daily plan

The problem

After a hospital stay, patients, especially older adults, go home with many new medicines and a dense summary they struggle to read. WHO has reported that only about half of patients on long-term therapies take them as prescribed. Misunderstood instructions and unsafe combinations are a preventable cause of harm, and many patients and caregivers do not read English comfortably.

Our solution

RxRosetta takes the medicine lines from a discharge paper (pasted text; photo or PDF transcription with a vision model is an experimental option) and builds a plain-language plan in English, Tamil or Hindi. Every check is done by a deterministic engine backed by a curated, auditable knowledge base, so no medical claim comes from a language model's memory. Uncertain or unknown medicines are flagged for a pharmacist rather than guessed.

Key features

Understands real formats	Indian 1-0-1 notation, brand names (Dolo, Ecosprin, Glycomet), abbreviations (BD, TDS, HS) and small spelling or OCR errors.
Daily plan	Morning, afternoon, evening and night timeline with food rules, the purpose of each medicine in simple words, and read-aloud audio.
Grounded safety checks	Interaction and duplicate-therapy warnings, each with a DailyMed link. Unknown or uncertain medicines are flagged: verify with your pharmacist.
Teach-back	A three-question quiz checks the patient actually understood. A low score alerts the caregiver.
Red-flag check-in	Rule-based triage returns stable, call your doctor today, or get emergency care now, using the patient's own medicines (for example, bleeding while on a blood thinner).
Caregiver feed	Live view of doses taken, quiz results and check-in alerts.

How we built it

Python and FastAPI backend, vanilla JavaScript front end with a calm, large-text design. Parser with fuzzy matching, an interaction engine, rule-based triage and a built-in evaluation harness. An experimental hook lets a vision model (Claude or Gemini) transcribe photos into the same pipeline. No real patient data is stored; all samples are synthetic.

Evaluation

On 40 seeded synthetic discharge notes (227 medicines) with brand names, typos and OCR-style character confusions, the rule-based parser reached 94.3% medicine recall, 100% precision, 100% dose accuracy, 100% schedule accuracy and 100% interaction recall on the medicines it found. The misses were badly garbled names; the system flags uncertainty instead of guessing. This is a synthetic baseline, not a clinical validation.

Impact and next steps

Aimed at patients and families who need clarity most, starting with Tamil, Hindi and English. Next: pharmacist-reviewed knowledge base expansion, live drug-label lookups, reliable photo transcription, clinician validation on de-identified real documents, and SMS or WhatsApp reminders.

Decision support only. Not medical advice. Team: [add names]. Code: [add GitHub link]. Demo: [add link].